AAE 571 Complex System Safety 3 credits

Commented [ZS1]: From previous version: get the new Rausand and Holland, has stuff for reliability modules including how to find reliability data https://onlinelibrary.wiley.com/doi/book/10.1002/9781119373940

1 Course Overview:

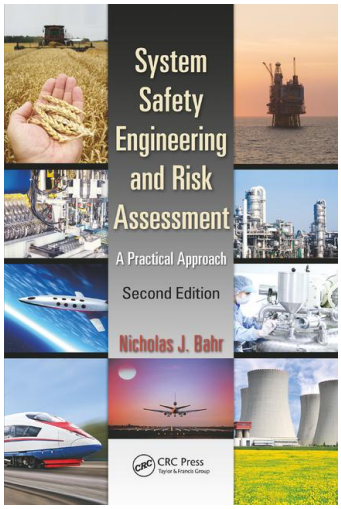
The prevention of complex system failures remains a critical challenge in engineering practice and research. Despite significant advancements in technology and methodology, accidents continue to occur. This course seeks to address the underlying reasons why accidents are so difficult to prevent despite decades of experience and to explore how contemporary theories in risk assessment and accident progression can inform safer system design and operation.

2 Learning Resources and Reference Text

Brightspace Learning Management System:

Access the course videos, materials, and resources via Purdue's BS learning management system. Begin with the Start Here tab, which describes how the course Brightspace is organized. We strongly suggest that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

Textbook:



Available online at no cost through Purdue Libraries.

3 Assignments

The course has team and individual assignments. You will have one team for all team assignments. We will have three main types of assignments:

3.1 Individual Weekly Reflections

Each week, we will have one or more readings, and ask you to prepare a response to the readings and class material.

3.2 Team Assignment 1: One Accident Case Study

Prepare a 15-minute video presentation on your accident, and a one-page summary. Your one-page summary should make appropriate use of text and graphics to convey efficiently the most important aspects of the accidents. See BS for examples.

3.3 Team Assignment 2: Team Term Project

- Apply the class material to a topic of your interest. I am open to a wide range of topics, please get in touch with me to discuss your ideas.
- Prepare a conference style paper
- Prepare a final presentation video (15 to 20 minutes, no more than 20 minutes).

4 Course Learning Objectives

- Describe several frameworks and theories for assessing and improving system safety
- Describe key elements of published system safety and risk assessment studies
- Describe different models used to model accidents.
- Apply different models to model accidents.
- List the steps in developing a system safety plan
- Recognize the extent of problems in system safety and risk assessment
- Describe the role of various systems and factors in creating safety and in causing errors and adverse events
- Discuss problems and issues in measuring and reporting safety
- Demonstrate knowledge of the basics of conducting an incident or accident investigation
- Design solutions to improve safety
- Discuss the ethical, legal, and regulatory implications related to safety
- Compare the different types of instruments/methods available to assess system safety
- Critique the use of commonly used system safety and risk assessment techniques in specific applications

5 Evaluation

All assignments, presentations, etc., must be submitted on BS or Gradescope as instructed. No email submissions will be accepted. Late submissions will be accepted only in exceptional circumstances. Make sure to communicate with me clearly and ASAP in case you anticipate a late submission, or have missed a submission.

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Your grade will be calculated as follows:

- Accident Case Study: 25%

- Weekly Reflections: 20%
- Team Term Project: 50%
- Final Project Peer Review: 5%

Your course grade will be based on the following grading scale:

Letter Grade	Percentage
A+	97-100%
A	92-96%
A-	90-91%
B+	87-89%
B	84-86%
B-	80-83%
C+	77-79%
C	74-76%
C-	70-73%

Letter Grade	Percentage
D+	67-69%
D	64-66%
D-	60-63%
F	0-59%